

ABRAR ZAHIN RAIHAN

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🌐 [Abrar Zahin Raihan](#)

🔗 [azraihaan](#)

EDUCATION

Bangladesh University of Engineering and Technology

Bachelor of Engineering in Computer Science and Engineering

Nov 2022 – Present

Dhaka, Bangladesh

Notre Dame College

College

Jul 2019– Dec 2021

Dhaka, Bangladesh

St. Joseph Higher Secondary School

High School

Jan 2016– Feb 2019

Dhaka, Bangladesh

SKILLS

Programming Languages: C, C++, Shell/Bash, SQL, Python, Java, TypeScript

Frameworks: PyTorch, Django, FastAPI, Express, Spring Boot, Next

Tools: GitHub Actions, Nginx., Prometheus, Grafana, k6, Docker, n8n

Clouds: Kubernetes, GCP, Azure, Terraform

Databases: PostgreSQL, MongoDB, Redis

Specialties: Machine Learning/Deep Learning, Gen AI, System Design, Data Engineering, CI/CD, Full Stack Development

EXPERIENCE

🌐 [SysModeler.ai](#)

Research Assistant (Remote)

November 2025 – February 2026

Canton, Michigan, United States

- **SysML Layout Optimization** Designed and implemented robust layout algorithms to generate structured, readable, and visually consistent SysML diagrams
- **Image-to-SysML Generation** Developed a pipeline to convert images into aesthetically refined and semantically accurate SysML representations

🌐 [Syslabo Corp](#)

SWE Intern (Remote)

Jul 2024 – Sep 2024

Tokyo, Japan

- **Organizational Hierarchy Chart Management System** Worked on the automation of the creation and tracking of syslabo's organizational hierarchy charts, replacing their current manual Excel-based system, for their digital workflows. Built a full-stack MERN application for this purpose.
- **Interactive UI/UX design & Japanese Localization** For the organizational chart application, developed a dynamic UI/UX with robust visualizations intuitive to syslabo's employees with Japanese language support using MaterialUI and GoJS.
- **System Design** Designed scalable database schemas and a monolithic architecture with RESTful APIs for efficient hierarchical data management.
- **Outstanding Performance Recognition** Received recognition for the best application architecture design among all interns and secured a full-time job offer for exceptional performance.
- [🔗 Internship Project](#)

PROJECTS

🌐 [CodeSprout](#)

| *Spring Boot, React, Python, ML, CV, Gen AI, Vector Database*

Jul 2024 – Sep 2024

- AI-powered coding education platform with gamified block-based programming for children
- Developed ML-powered chatbot for personalized learning recommendations and adaptive content
- Implemented automated project generation system based on individual learning progress
- Deployed using Java/Spring Boot microservices architecture with React frontend
- Integrated computer vision for real-time drawing-to-image generation

🌐 [DevOps Project \(BUET CSE Fest Hackathon 2024\)](#)

System Design, Azure, Kubernetes, Docker, Terraform, Prometheus, Grafana, k6, Express, Redis

Oct 2024

- Built a real time ticketing system using microservices for authentication and booking using Express.js and Redis caching
- Containerized services with Docker and K8s for automated scaling and load balancing
- Implemented CI/CD with GitHub Actions for zero-downtime deployments
- Set up monitoring with Prometheus/Grafana and k6 load testing
- Designed RESTful APIs with rate limiting and OTP verification mocking

HONORS & AWARDS

- [🌐 CUET ETE Televerse 1.0 : AIFication 2025 \(Datathon\) - First Runner-up](#) Worked on regional Bengali dialect speech to standard Bengali text conversion. **Nov 2025**
- [🌐 IEEE VIP Cup 2025 - Champions](#) Developed multimodal RGB-IR fusion models for real-time drone detection, tracking, and payload classification in surveillance videos. Presented our solution in ICIP 2025. **Jul 2025**
- [🌐 Poridhi.io and Brain Station 23 AI Engineering Hackathon - Champions](#) We built an intent-based product search with llama.cpp, optimizing for low-resource use and multilingual support. **Apr 2025**
- [🌐 BUET CSE Fest Hackathon 2024 - 2nd Runner-up](#) Microservices and DevOps Category **Oct 2024**
- [🌐 Therap JavaFest 2024 - Top 10 finalist](#) Worked on developing an AI coding tutor with gamified learning for kids, ranked top 10 among 1300+ participants **Sep 2024**
- [🌐 Code Samurai Hackathon 2024 - 10th Place](#) Engineered a waste management system for the DNCC (as proposed by BJIT), solved algorithmic challenges, and built a web app and a React Native mobile app to implement the system **Mar 2024**